Copernicus Global Land Operations
"Vegetation and Energy"
"CGLOPS-1"

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PRODUCT USER MANUAL

LAND SURFACE PHENOLOGY

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List of Acronyms

AD	Applicable document
AMPL	Seasonal amplitude
ATBD	Algorithm Theoretical Basis Document
B	Bias
BRDF	Bi-directional Reflectance Distribution Function
CGLOPS	Copernicus Global Land Operations
CGLS	Copernicus Global Land Service
CLMS	Copernicus Land Monitoring Service
COG	Cloud-Optimized GeoTIFF
DN	Digital Number
DVI	Difference Vegetation Index
EBF	Evergreen Broadleaf Forest
ECMWF	European Centre for Medium-Range Weather Forecasts
EEA	European Environment Agency
EPSG	European Petroleum Survey Group
ERA5	ECMWF Reanalysis v5
ESA	European Space Agency
EOSD	End of season date
EOSV	Vegetation index value at EOSD
EVAL	Evaluation newtork
GeoTIFF	Georeferenced Tag Image File Format
GPP	Gross Primary Production
HR-VPP	Copernicus High Resolution Vegetation Phenology and Productivity dataset
ICOS	Integrated Carbon Observation System of Europe
JRC	Joint Research Centre of European Commission
LAI	Leaf Area Index
LANDVAL	Land Validation network
LENGTH	Length of season
LSLOPE	Slope of the green-up period
LSP	Land Surface Phenology
MAD	Median Absolute Deviation
MAXD	Day of the maximum of season
MAXV	Maximum value of the season at MAXD
MD	Median Deviation
MDVI	Maximum DVI over a time period

MINV	Minimum value of season.
MOD	MODIS products based on TERRA data
MODIS	Moderate Resolution Imaging Spectroradiometer
MR-VPP	EEA Medium Resolution Vegetation Phenology and Productivity dataset
MuSLI	Multi-Source Land Imaging
NBAR	Nadir BRDF-Adjusted Reflectance
NDVI	Normalized Difference Vegetation Index
NEON	National Ecological Observatory Network
NIR	Near-Infrared
OLCI	Ocean and Land Colour Instrument
PEP725	Pan-European Phenology Project
PPI	Plant Phenology Index
PROBA-V	Project for On-Board Autonomy - Vegetation
PUM	Product User Manual
QA	Quality Assessment
QAR	Quality Assessment Report
QUIL	Quality Index Layer
R	Correlation coefficient
ReBeLS	Regularised BRDF inversion for Land Surface
RGB	Red-Green-Blue
RMSD	Root Mean Square Deviation
RSLOPE	Slope of the green-down period
S3	ESA Sentinel-3 satellite
SNR	Singal-to-Noise Ratio
SOSD	Start of season date
SOSV	Vegetation index value at SOSD
SPROD	Seasonal productivity
STD	Standard deviation
TIMESAT	a program for analysing time-series of satellite sensor data
TOC	Top-of-Canopy
TPROD	Total productivity
VIIRS	Visible Infrared Imaging Radiometer Suite
VNP	SNPP/VIIRS suite of products
VPP	Vegetation Phenology and Productivity
WGS	World Geodetic System

EXECUTIVE SUMMARY

The Copernicus Global Land Service (CGLS) is earmarked as a component of the Land service to operate "a multi-purpose service component" that provides a series of bio-geophysical products on the status and evolution of land surface at global scale. Production and delivery of the parameters take place in a timely manner and are complemented by the constitution of long-term time series.

The portfolio of the Copernicus Global Land Service contains a Land Surface Phenology (LSP) product at 300m spatial resolution. The global 300m LSP Version 1.0 (LSP 300m V1.0) product is derived from the smooth seasonal trajectories of the plant phenology index (PPI). The index is computed using 5-day Nadir Bidirectional Reflectance Distribution Function (BRDF)-adjusted Top-Of-Canopy (TOC) reflectance (NBAR) in red and near-infrared (NIR) bands of the Ocean and Land Colour Instrument (OLCI) onboard the Sentinel-3 (S3) platform.

The CGLS LSP 300m V1.0 product provides information on vegetation phenology and productivity (VPP) up to two growing seasons. Each season includes 13 parameters and is accompanied by a Quality Assessment (QA) layer to ensure data reliability.

This document provides a detailed product description of the CGLS LSP 300m V1.0 product. It includes a detailed overview of the retrieval methodology, and information on file naming, file format, product content, product characteristics, QA information, data policies and data access.

1 BACKGROUND OF THE DOCUMENT

1.1 SCOPE AND OBJECTIVES

The Product User Manual (PUM) is the primary document that users have to read before handling the data from the product. It gives an overview of the product characteristics, in terms of algorithm, technical characteristics, and main validation results. The key information is summarized in Table 1.

Table 1: Summary of key information

Input data The global LSP 300m V1.0 products are generated from Sentinel-3/OLCI Nadir BRDF-adjusted TOC reflectances.
Retrieval algorithm The processing consists of BRDF correction at 5-day intervals, Maximum Difference Vegetation Index (MDVI) estimation, PPI computation, and LSP retrieval.
File naming c_gls_LSP-<VPP>-<SEASON>_<DATE>_<AREA>_<SENSOR>_V<VERSION>
File format and product content Cloud optimized GeoTIFF format (COG)
Projection and grid information EPSG:4326 with WGS 84 datum and grid resolution of 0.00297619°
Spatial information Longitude [180°W – 180°E], Latitude [80°N – 65°S]
Temporal information Annual product with up to two growing seasons for each year.
Data policies Data access is free and open
Data access More details on data access are available on http://land.copernicus.eu/ .
Contact Scientific and Technical contact: https://land.copernicus.eu/en/contact-service-helpdesk

1.2 CONTENT OF THE DOCUMENT

This document is structured as follows:

- Chapter 2 summarizes the retrieval methodology
- Chapter 3 describes the technical properties of the product
- Chapter 4 reports the main results of the quality assessment
- Chapter 5 gives insights about product usage

The requirements are recalled in Annex.

1.3 RELATED DOCUMENTS

1.3.1 Applicable documents

AD1: Part 2: Technical specifications of Framework Service Contract – Operation of the bio-geophysical variables systematic monitoring of the Global Land Component of the Copernicus Land Service 'CGLOPS' JRC/2023/OP/0273, 19th April 2023.

Available at https://etendering.ted.europa.eu/cft/cft-display.html?cftId=13795

1.3.2 Input

Document ID	Descriptor
CGLOPS1_ATBD_LSP300m-V1	Algorithm Theoretical Basis Document of the global Land Surface Phenology 300m version 1
CGLOPS1_ATBD_BRDFCorrection300m-V1	Algorithm Theoretical Basis Document of BRDF correction version 1
CGLOPS1_ATBD_NDVI300m-V2	Algorithm Theoretical Basis Document of the NDVI 300m version 2 product
CGLOPS1_QAR_LSP300m_OLCI-V1	Quality assessment report of the version 1 global Land Surface Phenology 300m from Sentinel-3/OLCI
CGLOPS1_PQAD_I2.30	Product Quality Assurance Document "Vegetation & Energy".

1.3.3 External documents

Document ID	Descriptor
GCOS#245	The 2022 GCOS ECVs Requirements

Available online at https://library.wmo.int/records/item/58111-the-2022-gcos-ecvs-requirements-gcos-245?offset=1

2 ALGORITHM

2.1 OVERVIEW

Land Surface Phenology (LSP) is the term for land surface vegetation phenology estimated from remotely sensed data (White and Nemani 2006). It involves the analysis of time series of vegetation indices, which provide quantitative measures of green biomass and photosynthetic activity (Caparros-Santiago et al. 2021). LSP is an invaluable indicator in ecosystem studies and climate research (Jin et al. 2019; Piao et al. 2019; Yang et al. 2015; Richardson et al. 2013; Yu et al. 2010; Menzel et al. 2006; Schwartz et al. 2006).

CGLOPS LSP is developed from a vegetation index sensitive to green leaf area, the plant phenology index (PPI). This is a physically based vegetation index, developed from a simplified solution to the radiative transfer equation (Jin and Eklundh 2014). PPI is linearly related to green leaf area index (LAI), strongly correlated with gross primary productivity (GPP), and suffers minimal influence of snow seasonality (Jin et al. 2017; Jin et al. 2019). The index was used in the CLMS High-Resolution Vegetation Phenology and Productivity (HR-VPP) data product for estimation of phenology from Sentinel-2 data. The index was found to be superior to other indices based on relationships with different reference datasets, particularly GPP from ICOS eddy covariance data (Tian et al. 2021).

LSP parameters are generated from seasonal trajectories that undergo smoothing and gap-filling procedures, providing information on vegetation phenology and productivity (VPP). In CGLOPS, this is carried out using the TIMESAT software package (Eklundh and Jönsson 2017; Jönsson and Eklundh 2004). Data fitting is based on cubic smoothing splines. These smooth fits track seasonal and interannual vegetation index variations while reducing the influence of short-term unwanted variations, e.g. caused by residual cloud, or radiometric or geometric errors. The method interpolates data gaps in the time series of vegetation indices. The seasonal trajectories from the method are smoothly joined resulting in one continuous function describing the full time period, while capturing the interannual seasonal variations. Noise-affected pixel values due to cloud or snow are weighted down in the function fitting process, and the data are fitted to the upper envelope of data to account for negatively biased noise.

From the seasonal trajectories (data not stored as output in the CGLOPS product), 13 phenology and productivity parameters for up to two growing seasons per year are extracted (Figure 1).

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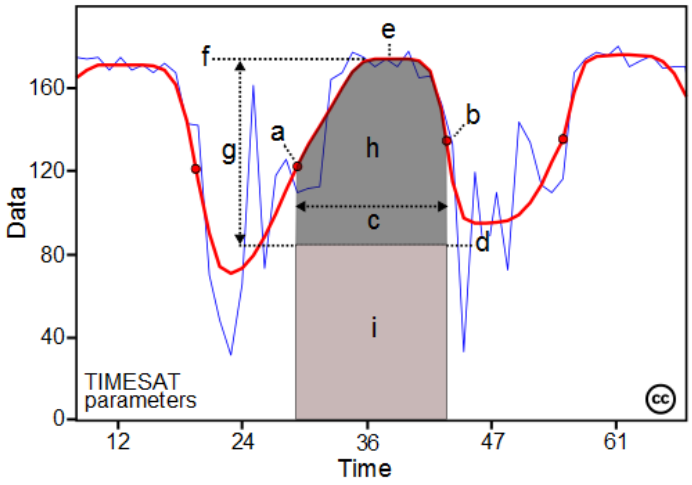


Figure 1. Examples of TIMESAT-derived phenology and productivity parameters: (a) start of season, (b) end of season, (c) length of season, (d) minimum value, (e) time of maximum, (f) maximum value, (g) amplitude, (h) seasonal productivity, (h+i) total productivity. Source: https://web.nateko.lu.se/timesat/timesat.asp.

The computation of start and end of seasons is based on defined threshold points on the smoothed seasonal trajectory.

2.2 INPUT DATA

2.2.1 Primary data

The input data for the LSP estimation are global 5-day PPI time series calculated from Sentinel-3/OLCI BRDF normalized top-of-canopy reflectances (TOC-r) in the red band (Oa08) and the near infrared (NIR) band (Oa17). The Sentinel-3/OLCI BRDF algorithm normalizes the TOC-r to the nadir viewing at 10:00AM local solar time, which is the Sentinel-3 overpass time. More details can be found in [CGLOPS1_ATBD_BRDFCorrection300m-V1] for the BRDF correction, and in section 2.3.1 for the computation of PPI.

Besides using the Oa08 and Oa17 bands of TOC-r BRDF data in the PPI calculation the product uses the quality information layer (QUIL) which provides information on the quality of the BRDF-correction. The QUIL indicates the following:

- whether the retrieval is based on observations in the 5-day time window or not,
- whether the inversion was successful,
- the quality of the BRDF inversion based on an estimation of the goodness of fit,
- whether the TOC-r is out of range (i.e. TOC-r < 0 or TOC-r >1).

The band selection in the PPI calculation differs from that of the CGLOPS NDVI calculation. For CGLOPS NDVI, multiple narrow bands are averaged to get broad-band red and NIR reflectances comparable to PROBA-V bands [CGLOPS1_ATBD_NDVI300m-V2]. In contrast, PPI for CGLOPS LSP is based on single narrow red and NIR bands with relatively high Signal-to-Noise ratio (SNR). This approach avoids reflectance uncertainties introduced by averaging bands due to varying atmospheric components (Campbell and Norman 1998).

2.2.2 Auxiliary data

Considering the dependency of remotely sensed land surface reflectance on illumination conditions, PPI stands out as the sole vegetation index that incorporates variations in light conditions. It achieves this by incorporating factors such as the diffuse fraction of solar illumination and the solar zenith angle during satellite passes into the PPI formulation. The solar diffuse fraction (*dc*) in CGLOPS PPI is derived from the European Centre for Medium-Range Weather Forecasts (ECMWF) Reanalysis v5 (ERA5) single levels dataset (https://cds.climate.copernicus.eu/cdsapp#!/dataset/reanalysis-era5-single-levels). Specifically, it is derived from the total sky direct solar radiation at the surface (*fdir*, J.m^{-2}) and the surface short-wave solar radiation (*ssrd*, J.m^{-2}) retrieved for the Sentinel-3 local overpass time of 10:00 AM. This calculation is expressed as $dc = 1 - fdir/ssrd$.

2.3 RETRIEVAL METHODOLOGY

2.3.1 Outline

Calculation of the LSP product is based on the following steps: (1) Data preparation: generating nadir BRDF-corrected TOC reflectance using ReBeLS NBAR retrieval algorithm and collecting ECMWF ERA5 diffuse fraction of solar radiation. (2) Generation of the plant phenology index (PPI) from Sentinel-3 OLCI TOC BRDF-corrected measurements at 300 m spatial resolution. (3) Processing of TIMESAT version 4.2 to derive smooth seasonal trajectories and phenology and productivity parameters (Figure 2).

For valid land pixels, PPI values are computed at five-day time step from BRDF-corrected Sentinel-3 OLCI red and NIR TOC values (NBAR). ERA5 data of diffuse to total radiation, and maximum difference vegetation index (MDVI) values from the entire NBAR time series from January 2023 to February 2024 are used in the computation (note that during the calibration phase PROBA-V NBAR test data were used from Jan. 2019 to Feb. 2020). The PPI data are fed through to TIMESAT 4.2 for smoothing, computation of phenology and productivity parameters, and derivation of QA flags.

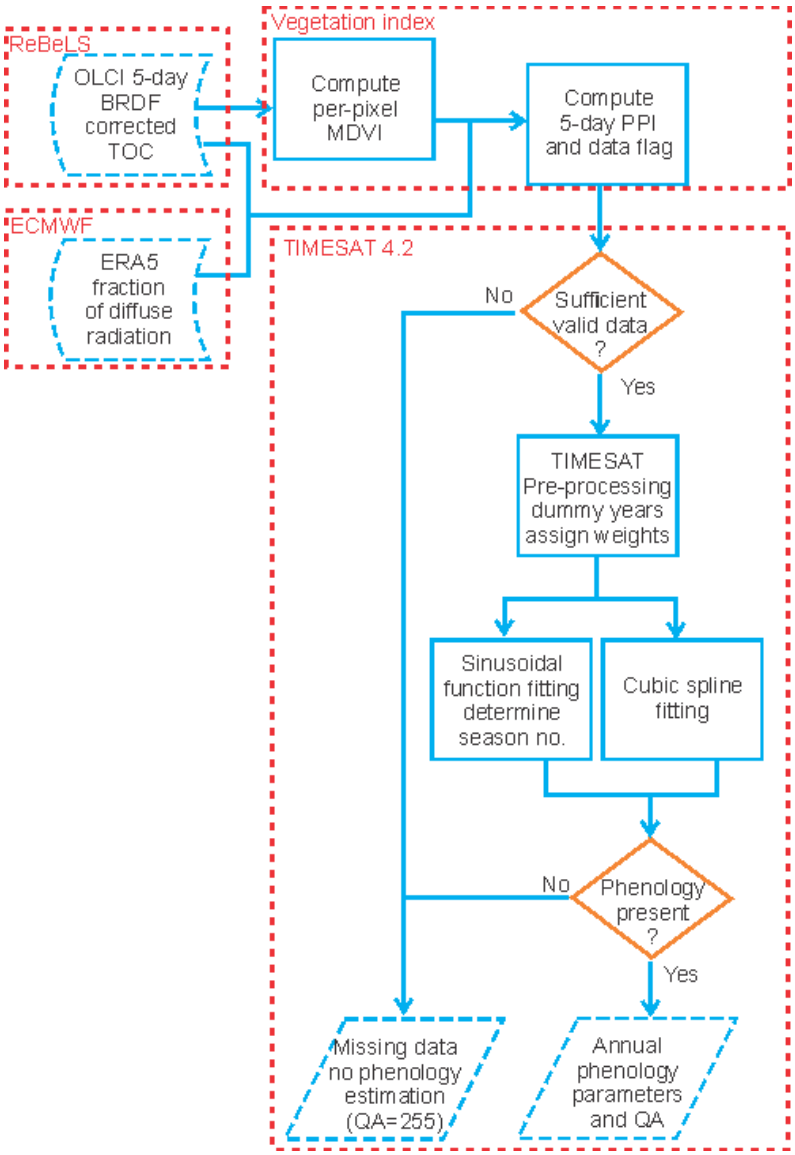


Figure 2. LSP processing overview.

2.3.2 Calculation of PPI

The calculation of PPI involves three main steps (Figure 3).

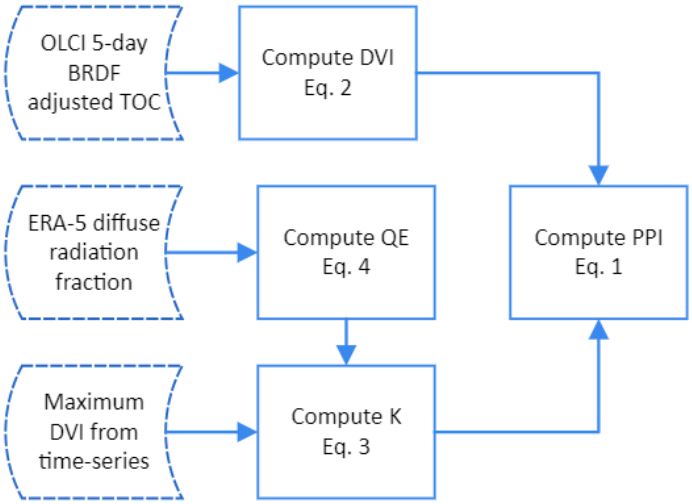


Figure 3. Calculation of PPI.

The formula of PPI (unit: $m^2.m^{-2}$) is:

$$PPI = -K \times \ln\left(\frac{MDVI - DVI}{MDVI - DVI_s}\right), \qquad Eq.1$$

where DVI is the difference vegetation index. DVI is a dimensionless index and is obtained by subtracting the red TOC reflectance from the near-infrared (NIR) TOC reflectance,

$$DVI = R_{NIR} - R_{RED}\,. \qquad Eq.\ 2$$

$MDVI$ is the temporal potential maximum DVI, which represents infinite leaf layers for a pixel with leaf traits that suit the environmental conditions of the location. $MDVI$ can be estimated from long-term satellite estimation, for example, MODIS Nadir BRDF-Adjusted Reflectance (NBAR) data of over 20 years, which are used in EEA Medium Resolution Vegetation Phenology and Productivity dataset (MR-VPP). The MDVI estimation from the maximum value of a moderately long (5+ years) time series of DVI works well in densely vegetated areas due to the asymptotic nature of vegetation spectral reflectance (Tucker 1977). However, in sparsely vegetated areas, the asymptotic MDVI level cannot be reached. To address this issue, a lower boundary value of 0.18 is assigned as the MDVI value for pixels below this threshold in the current version of the LSP product. This value represents the mode value of DVI across 297 PROBA-V tiles globally for 2018-2019, providing an enhanced spatial consistency of LSP productivity. DVI_s is used in Eq.1 to minimize influences of soil background on vegetation signals, and meanwhile to capture sparse vegetation areas such as drylands.

The steps for estimating $MDVI$ and DVI_s are: i) $MDVI = \max(DVI)$, from Sentinel-3 data from the available period (an alternative is to use PROBA-V data from 2014 to 2021). ii) $MDVI =$

$0.18, for\ MDVI < 0.18$. iii) $DVI_s = \min(0.09, MDVI_{scene}/4)$, and $MDVI_{scene} = \max(MDVI)$ over a scene.

The K factor in PPI is a gain factor given by

$$K = \frac{1}{4Q_E} \cdot \frac{1 + MDVI}{1 - MDVI}, \qquad Eq.3$$

where Q_E is the canopy leaf light extinction efficiency, the ratio of leaf cross-section area on the light beam to leaf geometric cross-section area, unit: $m^2.m^{-2}$ (Hapke 1993). Q_E is related to leaf inclination angle, solar angle, and the diffuse fraction of solar radiation:

$$Q_E = d_c + (1 - d_c) \cdot \frac{G}{\cos(\theta_i)}, \qquad Eq.4$$

where G is a geometric function of leaf angular distribution, set to 0.5, and d_c is an instantaneous diffuse fraction of solar radiation of the day and local overpass time of Sentinel-3, i.e. 10:00 AM. The d_c calculated from ERA5 at 0.25° x 0.25° resolution is downscaled to Sentinel-3 resolution (300 m) using bilinear interpolation. For the pixels without ERA5 data available, a backup algorithm for diffuse fraction at solar zenith angle θ_i for clear sky of standard atmosphere conditions is used (Jin and Eklundh 2014):

$$d_c = 0.0336 + \frac{0.0477}{\cos(\theta_i)}. \qquad Eq.5$$

The range of PPI is restricted between -1 and 5 to reduce the noise sensitivity in high PPI values. Values below -1 are set to -1 and values above 5 are set to 5.

2.3.3 Extraction of phenology and productivity parameters using TIMESAT

An overview of TIMESAT version 4.2 processing is shown in Figure 4.

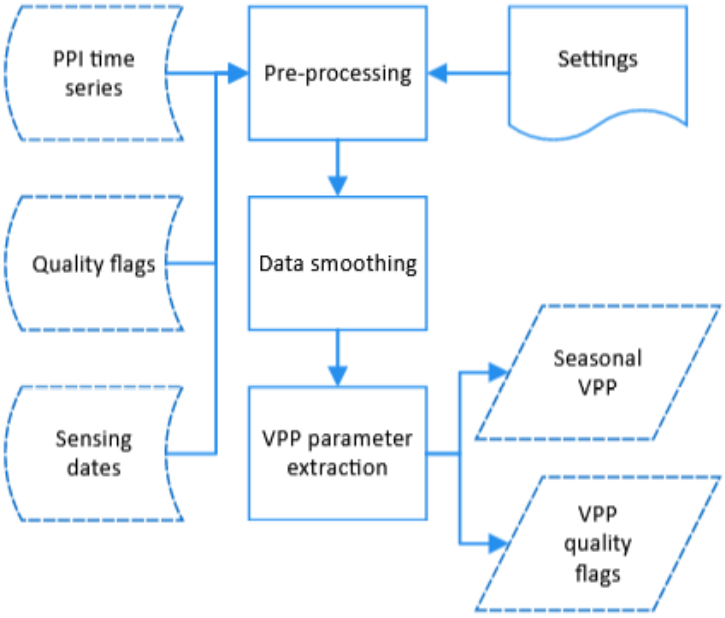


Figure 4. TIMESAT processing overview.

2.3.3.1 Pre-processing

The input PPI data for TIMESAT may include some noise and temporal gaps. The pre-processing has four purposes: (1) to transfer the input imagery data (PPI and PPI quality flags) to pixel-oriented time series for the later processing; (2) to fill large data gaps exceeding 91 continuous days with a predefined base level; (3) to preliminarily determine whether a time series needs to be excluded from processing; and (4) to make a preliminary analysis of seasonality (number of potential seasons) and to establish the rough timing of each coarse season.

The 91-day gap detection and filling mainly address the high-latitude winter seasons that are masked out in the BRDF processing [CGLOPS1_ATBD_BRDFCorrection300m-V1]. The base level is determined as the 5th percentile of the good quality PPI observations for the complete time period. The weights of PPI values in a time series are set based on PPI data quality flags. Furthermore, weights of PPI observations that are lower than the base level are set to 0. The base level is used for filling long-term data gaps and assisting the cubic spline fitting. TIMESAT will detect if there are enough good-quality data in the time series, see Section 2.3.2.6 for details. This step is to ensure the operation of fitting the cubic spline functions.

Pixel-wise sensing dates of Sentinel-3 reflectance values are converted to a sequential time vector, starting from the first day of the first year in the input period, e.g., 1 corresponding to 2023-01-01 if the input starts from 2023. To extract the phenological information of the first and last seasons from

the time series, TIMESAT 4.2 extends the length of the time series by duplicating the data of the first and last years.

2.3.3.2 Daily PPI construction

The PPI time series is fitted with a cubic smoothing spline $S_p(\mathrm{t})$ (Craven and Wahba 1978), which minimizes the value of a criterion function C_p

$$C_p = \sum_{i=1}^{n} \left\{ w_i \left[y_i - S_p(t_i) \right]^2 + p \int_{-\infty}^{+\infty} \left| S_p''(t) \right|^2 dt \right\}, \qquad Eq.6$$

where $t_i, i = 1, \ldots, n$ is time vector and y_i are corresponding PPI values. Each point in the time series is associated with an input weight w_i. The smoothing parameter p controls the shape of the spline, varying from an exactly interpolating spline ($p = 0$) to a straight line ($p \to \infty$). The p value is set to 1,000, which can reduce the impact of noise while preserving sufficient local variation.

2.3.3.3 Extraction of phenology and productivity parameters

An initial pixel-wise estimate of the growing seasons' start and stop dates ("coarse seasons") is determined by applying a sinusoidal function fitting

$$f(t) = c_1 + c_2 \sin(\omega t) + c_3 \cos(\omega t) + c_4 \sin(2\omega t) + c_5 \cos(2\omega t), \qquad Eq.7$$

where $\omega = 2\pi/n$, and n is the number of points, $f(t)$ is the corresponding PPI value at time *t*. The coarse seasons are further used for defining potential seasons.

From the reconstructed daily PPI time series, vegetation phenology and productivity (VPP) parameters are extracted, checked, and assigned to corresponding years where the peak is located (Figure 5). Thirteen VPP parameters and one quality layer for the 13 seasonal VPP parameters are generated from the processing, for delivery to users (Figure 1; Table 2).

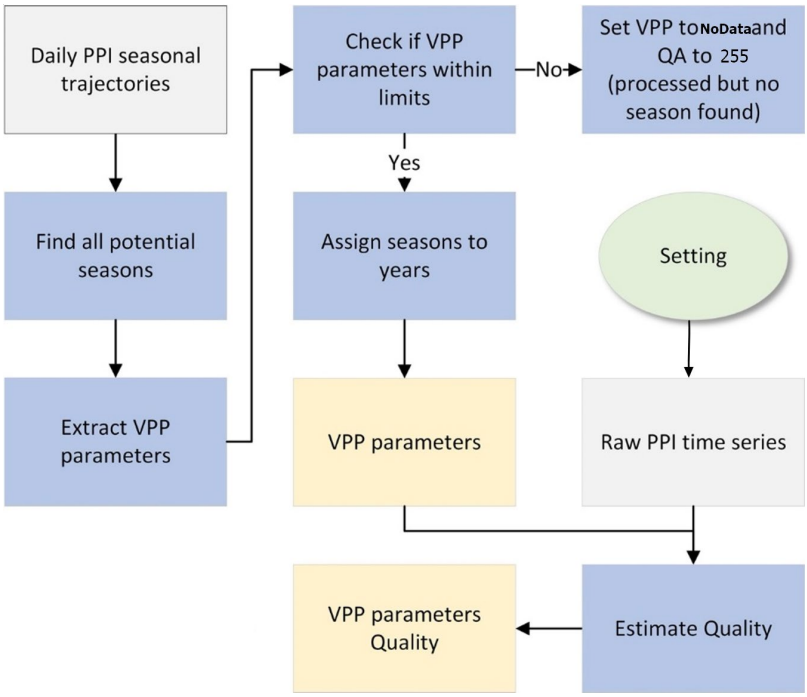


Figure 5. TIMESAT module for extracting phenological parameters.

2.3.3.4 Defining growing season thresholds

TIMESAT determines the timing of start of season (SOSD) and end of season (EOSD) by applying a threshold value in relation to the seasonal amplitude. To be consistent with the thresholds setting in both MR-VPP and HR-VPP products, 25% is used for SOSD and 15% for EOSD.

2.3.3.5 Assigning seasons to years

Seasons may traverse years, thus a method for assigning them to the correct year is necessary. Seasons are assigned to years based on the season peak timing, e.g., season a-A-b is assigned to year 2018 and season c-C-d-D-e is assigned to year 2019 in Figure 6. Annually, up to two seasons with the largest seasonal peaks are provided to users. The two seasons in a year are present chronologically, irrespective of their relative size. Thus, the main season may occur in Season 2 and a minor season in Season 1, or vice versa.

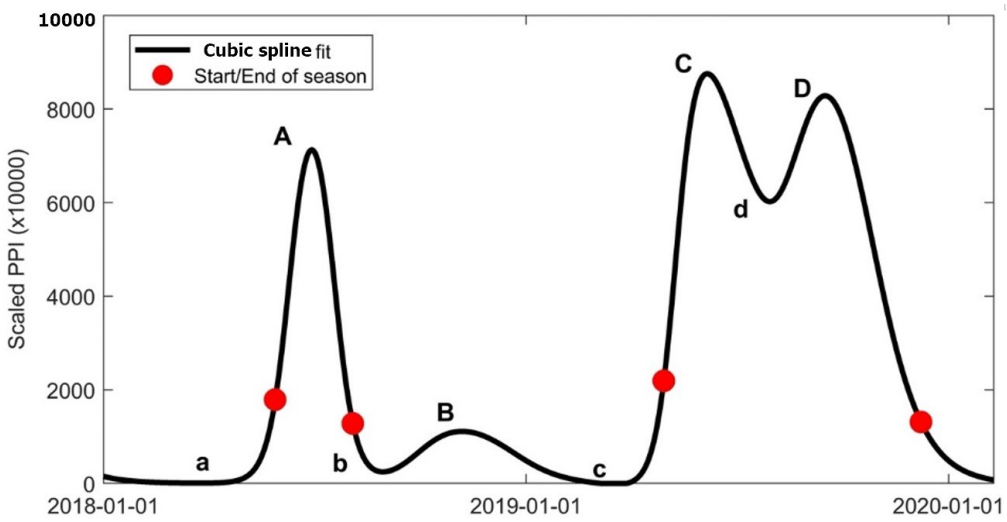


Figure 6. Example of the determination of seasons. The points, A, B, C, and D, are the peaks of coarse seasons, and the points, a, b, c, and d, are the corresponding local minima of coarse season. Red circles represent the start and end of the seasons.

2.3.3.6 Defining thresholds for omitting seasons and regions

In three cases, the LSP are set to nodata-values (with QA=255) and the TIMESAT processing is omitted:

1. If the number of valid input data points throughout the entire time series is less than 3, or 1% of the total number of the data points. This primarily applies to areas lacking valid observations.
2. If the first-order PPI difference of the time series is too small, with fewer than 5 points exhibiting significant differences (>1e10^{-6}). This indicates weak seasonality, making it difficult to precisely determine phenological parameters, and mainly occurs in areas with evergreen vegetation.
3. If the number of points with PPI values above 2% of the peak value is less than 5 points. This primarily omits inland water bodies with low PPI values occasionally exhibiting few extreme values, possibly caused by noise. Few pixels at the edge of waterbodies pass the screening but generate negative total productivity. The seasons for these pixels are omitted after processing.

2.3.4 Related and previous applications

This is the first version of CGLOPS LSP. It is modelled on similar principles as developed for the Copernicus Land Monitoring Service *High Resolution Vegetation Phenology and Productivity (HR-VPP)* (https://land.copernicus.eu/pan-european/biophysical-parameters/high-resolution-vegetation-

phenology-and-productivity), however at 300 m resolution based on S3 OLCI data and with global coverage. Processing methods have been adapted to the nature of the data from Sentinel-3 OLCI.

2.4 LIMITATIONS OF THE PRODUCT

PPI is a physically based index responding to green leaf area index (LAI) variations, but it is based on some simplified assumptions. It is primarily designed for homogeneous canopies, which may not hold true for open forests and bare areas. Its assumption about spherical (uniform) leaf angular distribution may not apply to certain crops. Reflectance retrieval assumes constant soil reflectance and relies on short time series data, leading to noise sensitivity, especially during peak growing seasons. PPI is limited to a maximum value of 5.0 globally and is smoothed using TIMESAT. The index is sensitive to BRDF effects, which means that errors in the BRDF processing may affect its performance.

In areas with sparse vegetation, attaining a maximum DVI value indicative of a sufficiently dense leaf layer is often unattainable, making it impossible to estimate the correct MDVI value from a long DVI time series. Therefore, we assign an empirical MDVI value of 0.18 to pixels with MDVI values below this threshold. However, it is important to note that this default MDVI value may require adjustment to better align with specific environmental conditions associated with various land cover types. In crop rotation systems, the assumption of a fixed MDVI level is challenged, potentially resulting in an underestimated seasonal trajectory for less dense crops.

Winter data gaps are filled using a base-level value, potentially hindering the detection of winter crop seasons. Conversely, the CGLOPS NBAR algorithm may sometimes generate inaccurate reflectance values for raw data gap periods, leading to the detection of false seasons. Considering these biases, we assigned a poor quality (QA="11") to phenology metrics with fewer than 3 valid data points based on NBAR data QUIL information. Caution is advised when utilizing metrics with low QA.

Additionally, since the outputs are limited to two seasons annually, minor seasons may be overlooked if there are more than two vegetation cycles per year, such as multiple mowing/grazing events in managed grasslands, and multi-season croplands exceeding two cycles.

Having access to a sufficiently long time series of land surface remote sensing data is essential for mitigating biases in CGLOPS global Land Surface Phenology (LSP) due to the diverse vegetation types and seasonal variations worldwide. However, utilizing short time series of CGLOPS NBAR data and focusing on a limited test area could potentially compromise the quality of the final LSP product, highlighting the constraints in LSP estimation caused by insufficient time series NBAR data. This limitation is expected to be alleviated with the ongoing acquisition of Sentinel-3 OLCI data and BRDF correction. Nevertheless, caution should be exercised when utilizing phenology data from the first and last years of the CGLOPS LSP product, particularly for the SOSD estimates from the

preceding year and the EOSD estimates from the following year, as these involve extrapolation to compensate for incomplete seasonal data.

2.5 DIFFERENCES WITH PREVIOUS VERSIONS

This is the first version of the product.

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3 PRODUCT DESCRIPTION

3.1 FILE NAMING

The global 300m **Land Surface Phenology (LSP300)** V1.0. product follows the naming standard:

c_gls_<PROD>-<VPP>-<SEASON>_<DATE>_<AREA>_<SENSOR>_V<VERSION>where

- c_gls stands for Copernicus Global Land Service
- <PROD> is LSP300 for the Land Surface Phenology at 300m resolution
- <VPP> denotes the vegetation phenology and productivity parameter (Table 2)
- <SEASON> indicates which season is listed, the values being S1 for season 1 and S2 for season 2.
- <DATE> gives the temporal location of the file. The date is formatted as YYYY0101000 where YYYY denotes the year. One product is delivered per year.
- <AREA> gives the spatial coverage of the file. In our case, <AREA> is GLOBE, short name for global.
- <SENSOR> gives the name of the sensor (here OLCI) used to retrieve the product.
- <VERSION> shows the processing line version used to generate these LSP300 products. The version denoted as M.m.r (e.g. 2.1.1), with 'M' representing the major version (e.g. V2), 'm' the minor version (starting from 0) and 'r' the production run number (starting from 1) (Table 2)

Table 2: Explanation in version numbering and recommendations for using efficiently the products

Versions	Differences	Recommendations
Major	Significant change to the algorithm.	Do not mix various major versions in the same application, unless it is otherwise stated.
Minor	Minor changes in the algorithm	Can be mixed in the same application, but require attention or modest modifications
Run	When a reprocessing is necessary to fix bugs and minor issues. Later run automatically replaces former.	Consider it as a drop-in replacement

3.2 FILE FORMAT

The global 300m LSP V1.0 product is distributed in Cloud Optimised GeoTIFF (COG) raster files with metadata attributes compliant with version 1.6 of the Climate & Forecast conventions (CF V1.6). Each COG raster file contains one data layer. A total of two seasons are identified and 14 vegetation parameters are available per season.

3.3 PRODUCT CONTENT

3.3.1 VPP Data Files

The physical range and scaling factors of the different vegetation phenology and productivity (VPP) parameters are given in Table 3. The physical values (PhyVal) are retrieved from the Digital Numbers (DN) through the following equation:

$$PhyVal = DN * scale + offset \qquad Eq.7$$

The physical value range of the Start-Of-Season Date (SOSD) is [-365, 365]. SOSD < 0 indicates that the peak of growing season is in the current year and the start of the season is located in the previous year.

The physical value range of the End-Of-Season Date (EOSD) is [0, 730]. EOSD > 365 indicates that the peak of growing season is in the current year and the end of the season is located in the next year.

It is recommended to use the VPP parameters together with their Quality Assurance (QA) flag (Section 3.3.2). If only one-year data is used as input, all phenology metrics below 0 and above (365+60) are not reliable (QA=3 poor quality - Table 4). If there are multiple-year data as input, these metrics might be reliable depending on how QA is.

Table 3. Vegetation Phenology and Productivity parameters

No.	Name	Data type	Scale, offset	Data range	Fill value	Description	Unit
1	SOSD	INT16	1, 0	[-365, 365]	-9999	Day of start-of-season	Day-Of-Year
2	EOSD	INT16	1, 0	[0, 730]	-9999	Day of end-of-season	
3	MAXD	INT16	1, 0	[0, 366]	-9999	Day of maximum-of-season	
4	SOSV	INT16	0.001, 0	[0, 5000]	-9999	Vegetation index value at SOSD	PPI unit $m^2 \times m^{-2}$
5	EOSV	INT16	0.001, 0	[0, 5000]	-9999	Vegetation index value at EOSD	
6	MINV	INT16	0.001, 0	[0, 5000]	-9999	Average vegetation index value of minima on left and right sides of each season	
7	MAXV	INT16	0.001, 0	[0, 5000]	-9999	Vegetation index value at MAXD	
8	AMPL	INT16	0.001, 0	[0, 5000]	-9999	Season amplitude (MAXV – MINV)	
9	LENGTH	INT16	1, 0	[0, 730]	-9999	Length of Season (number of days between start and end)	Day
10	LSLOPE	INT16	0.001, 0	[-1000, 1000]	-9999	Slope of the green-up period	$m^2 \times m^{-2} \times day^{-1}$
11	RSLOPE	INT16	0.001, 0	[-1000, 1000]	-9999	Slope of the green-down period (absolute value of decreasing rate)	
12	TPROD	INT16	1, 0	[0, 2000]	-9999	Total productivity. Growing season integral computed as the sum of all daily values between SOSD and EOSD.	$m^2 \times m^{-2} \times day$
13	SPROD	INT16	1, 0	[0, 2000]	-9999	Seasonal productivity. Growing season integral computed as sum of all daily values minus their base level value.	
Aux	QA	UINT8	-	-	255	Quality flag	-

3.3.2 VPP QA

VPP Quality Assurance (QA) is assigned based on both the average of PPI data quality and the total number of valid observations in each of phenological phases: green-up, green peak, and green-down of the growing season. They are defined as the left 20% – left 80%, left 80% – right 80%, and right 80% – right 20% of the season amplitude respectively (Figure 7). The overall QA is determined based on the comprehensive evaluation of the quality across all three phases (Table 4).

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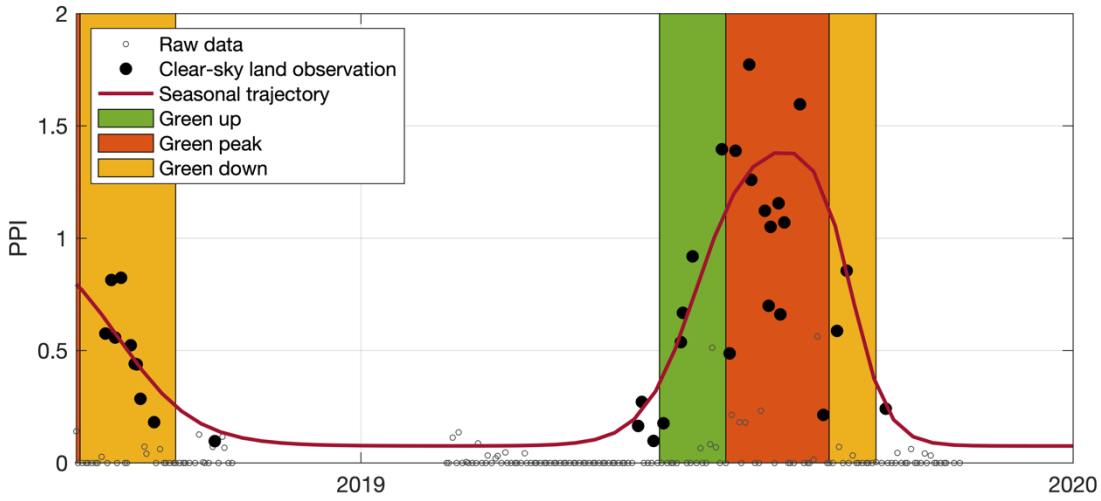


Figure 7. Example of VPP parameters QA.

Eight-bit (1 byte) QA values are generated as specified in Figure 8, and the details of bit values and descriptions are listed in Table 4. The QA flag is written to one output file for each season.

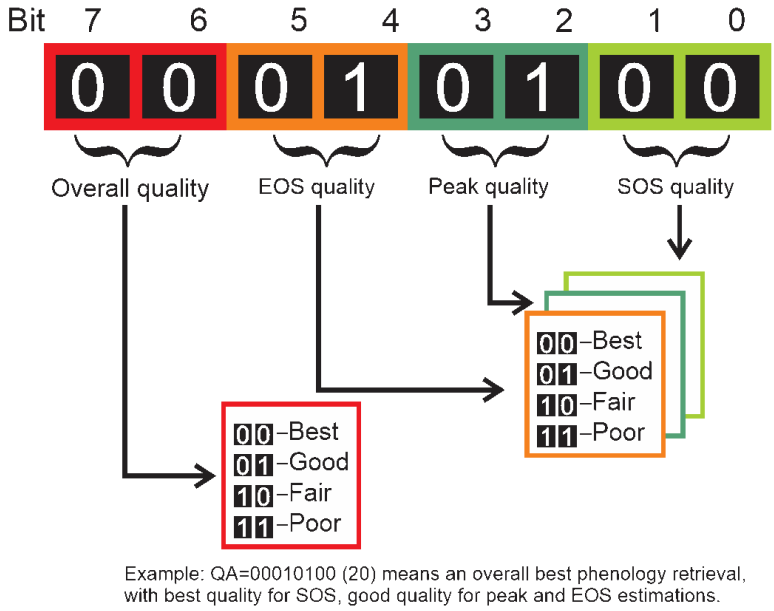


Figure 8. Details of 8-bit quality flags for phenology retrieval over a season.

Table 4. LSP QA flags and descriptions of phenology quality.

QA byte	Quality	Description
SOS/Peak/EOS quality (two-bit values)		
0	Best	More than 2 valid observations and average best PPI quality
1	Good	More than 2 valid observations and average good PPI quality
2	Fair	More than 2 valid observations and average fair PPI quality
3	Poor	Average poor PPI data quality, or fewer than 3 observations, or more than > 24 observations*
Overall quality (two-bit values)		
0	Best	No poor quality in three phenology phases
1	Good	Maximum one phase with poor quality
2	Fair	Maximum two phases with poor quality
3	Poor	All three phases with poor quality**
Summary of QA (byte values)		
QA <127	Good	Overall good quality, no more than one poor results in three phases
QA >127	Poor	Overall poor quality, more than one poor results in three phases
QA=255	Failure	No input, or no phenology estimation, or all poor estimates

Notes:
* Includes three situations: 1) fewer than 3 valid observations, 2) phenology phase shorter than 10 days, and 3) phenology phase longer than 4 months.
** Overlaps QA=255 for situations of no inputs, or no phenology estimation. In any case, the LSP retrieval is considered a failure.

3.3.3 Metadata

The images include metadata on 2 levels: file and layer level. The metadata in Table 5 is included on file level while the attributes in Table 6 are provided at layer level.

Metadata follows the CF-1.6 convention for NetCDF files. The COG standard names are used instead of CF-1.6 NetCDF specific names for those attributes that have a standardized COG equivalent name. Whenever this is the case, it is indicated in the description column of the following tables.

Table 5: Description of file attributes

Attribute name	Description	Example(s)
Conventions	Version of the CF-Conventions used	CF-1.6
title	A description of the contents of the file	Yearly Land Surface Phenology 333M: GLOBE 2023
institution	The name of the institution that produced the product	VITO NV
source	The method of production of the original data	Derived from EO satellite imagery
history	A global attribute for an audit trail. One line, including date in ISO-8601 format, for each invocation of a program that has modified the dataset.	2024-03-22: Processing line LSP
references	Published or web-based references that describe the data or methods used to produce it.	https://land.copernicus.eu/en/products/vegetation
archive_facility	Specifies the name of the institution that archives the product	VITO
product_version	Version of the product (VM.m.r)	V1.0.1
time_coverage_start	Start date and time of the total coverage of the data for the product.	2023-01-01T00:00:00Z
time_coverage_end	End date and time of the total coverage of the data for the product.	2024-02-29T23:59:59Z
platform	Name(s) of the orbiting platform(s)	Sentinel 3
sensor	Name(s) of the sensor(s) used	OLCI
identifier	Unique identifier for the product	urn:cgls:global:lsp300_v1_333m:LSP300-AMPL-S1_202301010000_GLOBE_OLCI_V1.0.1
parent_identifier	Identifier of the product (time series) for the product in Copernicus Global Land Service metadata catalogue.	urn:cgls:global:lsp300_v1_333m
long_name	Extended product name	Land Surface Phenology
orbit_type	Orbit type of the orbiting platform(s)	LEO

Attribute name	Description	Example(s)
processing_level	Product processing level	L3
processing_mode	Processing mode used when generating the product (Near-Real Time, Consolidated, Offline or Reprocessing)	Offline
copyright	Text to be used by users when referring to the data source of this product in publications (copyright notice)	Copernicus Service information 2024

Table 6: Description of layer attributes

Attribute	Description	Examples for SOSV layer
CLASS	Dataset type	
short_name	A shortened, non-standardized name.	SOSV
long_name	A descriptive name that indicates a variable's content. This name is not standardized. Required when a standard name is not available.	Start-Of-Season Value
scale	Multiplication factor for the variable's contents that must be applied in order to obtain the real values. Omit in case the scale is 1. Tiff compliant version of CF-1.6 scale_factor.	0.001
offset	Number to be added to the variable's contents (after applying scale_factor) that must be applied in order to obtain the real values. Omit for offset 0. Tiff compliant version of CF-1.6 add_offset.	
units	Units of the variable's content, taken from Unidata's udunits library. Empty or omitted for dimensionless variables. Fractions should be indicated by scale.	m^2.m^-2
valid_range	Smallest and largest values for the variable. Missing data is to be represented by one or several values outside of this range.	0, 5000
missing_value	Single value used to represent missing or undefined data, for applications following older versions of the standards. Identical to the COG NoData Value of the layer. Value must be outside of valid_range.	-9999
flag_values	Provides a list of the flag values. Used in conjunction with flag_meanings and optional flag_masks. Omit in case no flags are specified	-
flag_meanings	Descriptive words or phrases for each flag value. Omit in case no flags are specified	-
flag_masks	Provides a list of flag masks to group multiple bits together. Omit in case no flags are specified or no bit grouping is done.	-
Season	Non-standardized number indicating season index, either 1 or 2.	1

3.4 PRODUCT CHARACTERISTICS

3.4.1 Projection and Grid Information

The product is displayed in a regular latitude/longitude grid (plate carrée) with the ellipsoid WGS 1984 (Terrestrial radius=6378km). The resolution of the grid is 1/336°.

The reference is the centre of the pixel. It means that the longitude of the upper left corner of the pixel is (pixel_longitude – angular_resolution/2.)

3.4.2 Spatial Information

The global 300m Land Surface Phenology Version 1.0 product is provided from longitude 180°E to 180°W and latitude 80°N to 65°S.

3.4.3 Temporal Information

The global 300m Land Surface Phenology Version 1.0 product is produced annually.
The temporal coverage of the global 2023 LSP300 product spans from 1st January 2023 to 29th February 2024.
The temporal coverage of the global 2024 LST300 products spans from 1st January 2023 to 28th February 2025.

3.5 DATA POLICIES

The Copernicus component is governed by Regulation (EU) No 2021/696 of the European Parliament and of the Council of 28 April 2021 establishing the Union Space Programme and the European Union Agency for the Space Programme.

The Copernicus land monitoring products and services are made available on a principle of **full, open and free access**, as established by the Commission Delegated Regulation (EU) No 1159/2013 of 12 July 2013.

How to cite and reference the products and services of the Copernicus Land Monitoring Service is explained here: https://land.copernicus.eu/en/data-policy.

3.6 ACCESS AND CONTACTS

The global 300m LSP V1.0 product is available through the Copernicus Land Monitoring Service website at the address: https://land.copernicus.eu/ with contact information (helpdesk) on https://land.copernicus.eu/en/contact-service-helpdesk.

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4 QUALITY ASSESSMENT

The quality assessment of CGLS OLCI LSP products was conducted for year 2023 [CGLOPS1_QAR_LSP300m_OLCI-V1]. The methodology is described in the CGLS Product Quality Assurance Document [CGLOPS1_PQAD_I2.30]. Two main validation approaches are defined: direct validation (i.e., comparison of satellite products with in situ observations) and indirect validation (or product intercomparison). Several criteria of performance wer analyzed, including the visualization of the product content, and the evaluation of the product completeness, spatial consistency and error (accuracy, precision and uncertainty).

The direct validation of SOSD and EOSD was performed with PEP725, TEMPO and NEON datasets, which provide phenological observations. PhenoCam V3 dataset, which infers phenometrics from digital cameras, was also used for direct comparison. The product intercomparison with global satellite-based products includes the qualitative comparison of SOSD/EOSD with VIIRS phenometrics (VNP22Q2 V1 for previous 2022 year due to unavailability of 2023 when QAR was performed) over a latitudinal transect and quantitative comparisons of TPROD/SPROD with CGLS OLCI and MODIS (MOD17A2HGF V6.1) GPP products over LANDVAL-2 network of sites. Finally, the comparison over Europe with equivalent MR-VPP and HR-VPP parameters is also performed over different sampling areas.

Main results for each criterion are:

Product content

- Reliable values were found for all CGLS OLCI LSP layers. However, OLCI products display unexpected seasons over few desert areas (e.g., Sahara and Arabia deserts).

Product completeness

- OLCI LSP showed similar overall completeness (75% of valid data for 2023) than VIIRS. OLCI LSP provides improved completeness over tropical areas but unexpected values over few desert areas.
- Considering the quality flag, less than 5% of OLCI data is filtered out (i.e., removing overall poor quality pixels, QA > 127). VIIRS quality flag is more restrictive, filtering out more than 20% of data.
- All products provided very few valid cases for season 2 (around 2%-4%).

Spatial consistency

- The analysis of OLCI LSP maps at full resolution over subcontinental regions and areas of interest showed consistent values in most cases (except over few desert areas, with unrealistic seasons).

- The maps of residuals of CGLS OLCI LSP versus MR-VPP over Europe showed around 60% of pixels within threshold requirement for SOSD and 35% for EOSD. Maps of differences showed delayed CGLS SOSD in most cases and large negative bias for EOSD mainly over Eastern Europe, which translates in shorter season in case of CGLS OLCI LSP.
- CGLS OLCI LSP and MR-VPP productivity layers showed residuals typically within ±40 $m^2.m^{-2}$.days, with main spatial discrepancies over southern sparsely vegetated regions. CGLS OLCI LSP tends to provide higher productivity than MR-VPP, even when the season is shorter, as the OLCI PPI reaches higher maximum values than MR-VPP.
- CGLS OLCI LSP products (SOSD, EOSD, SPROD and TPROD layers) were also spatially consistent with HR-VPP over selected areas of interest in Europe, showing similar spatial patterns for most cases.
- The analysis of SOSD and EOSD variations per latitude (latitudinal transect from Norway to South Africa) showed the expected trend and consistent values than equivalent VIIRS layers, except for desert areas and sparsely vegetated areas (around -25º latitude).
- The spatial variability over homogeneous areas revealed good precision for SOSD (STD typically $\leqslant$ 4%, MAD $\leqslant$ 3%) and for EOSD (STD and MAD $\leqslant$ 9%) over most of the evaluated areas, except for the EBF area where larger variability is found.
- The spatial variability of TPROD/SPROD was higher, with STD and MAD typically between 6% and 25%.

Error evaluation (direct validation)

- The comparison of CGLS OLCI LSP SOSD and EOSD versus ground measurements of PEP725, TEMPO and NEON datasets showed:
    - For SOSD, bias close to zero or slight delay (up to 5 days) was found in the comparison with ground phenophases related to the first stages of the growing season (e.g., 1st leaves separated (PEP725), germination (TEMPO) and breaking leaves (NEON)). As compared to phenophases related to more developed leaf growth (i.e., leaf development (TEMPO) and leaf unfolding ≥50% (PEP725)) negative bias (i.e., anticipated SOSD) between -7 and -12 days was found. These results suggest a good accuracy of the OLCI LSP SOSD, which represents a date between the first emerging buds/leaves and a more advance stage of leaf development. Better accuracy that MR-VPP was generally found for comparison with both PEP725 and TEMPO.
    - For EOSD, the bias was typically negative (i.e., anticipated end of season), and CGLS EOSD is typically more related to colouration stages (PEP725 and NEON) or beginning of dormancy (TEMPO) (bias ranging from -14 days to 4 days) than to the fall of leaves (up to -33 days). MR-VPP showed similar accuracy.
    - RMSD was typically ranging between 20 and 30 days for SOSD and from 20 to 40 days in case of EOSD.

  - The percentage of cases CGLS OLCI LSP within goal (i.e., 5 days) requirements typically ranges between 10% and 20% in case of SOSD (25%-40% for threshold, i.e., 10 days) and between 15% and 20% in case of EOSD (30%-45% for threshold).
- The results of the comparison with near-ground-based PhenoCam V3 were:
  - OLCI LSP SOSD better agreed in terms of accuracy and uncertainty with PhenoCam start of season defined at 50% threshold (Bias = 3 days, RMSD = 29 days) than compared with PhenoCam at 25% (Bias = 12 days, RMSD = 23 days), which indicates a delay of the SOSD in OLCI LSP as compared to PhenoCam phenology estimates (based on green chromatic coordinates).
  - For EOSD, the best agreement in terms of bias and uncertainty was found with PhenoCam defined at 25% (Bias = 0, MD = 2, RMSD = 22 days). Negative bias (Bias = -11 days) was found compared with PhenoCam defined at 10%.
  - GCLS OLCI LSP tends to provide shorter seasons as compared with near-ground-based datasets

Error evaluation (product intercomparison)

- The comparison of CGLS OLCI LSP with MR-VPP equivalent layers over Europe (EVAL sampling) showed:
  - Shorter season of CGLS, with delayed SOSD (B = 18 days) respect to MR-VPP and similar EOSD (B = -2 days). RMSD of 22 and 23 days for SOSD and EOSD.
  - CGLS TPROD/SPROD overestimated equivalent MR-VPP similar layers (B and MD around 25%) despite the shorter seasons, which is explained by the larger maximum values of PPI.
- The comparison of CGLS OLCI LSP with HR-VPP equivalent layers over Europe (EVAL sampling) showed:
  - Shorter season of CGLS as a result of delayed SOSD (B = 14 days) of CGLS respect to HR-VPP and slightly delayed EOSD (B = 6 days). RMSD of 21 and 24 days for SOSD and EOSD.
  - CGLS TPROD and SPROD overestimated (B around 30%) equivalent HR-VPP productivity layers, explained by the larger maximum CGLS PPI values.
- Comparisons of CGLS OLCI LSP TPROD and SPROD with other productivity GPP products showed consistent values: $R \geq 0.76$ with CGLS OLCI GPP and $R \geq 0.67$ with MODIS GPP.

Summary and conclusions

In summary, CGLS OLCI LSP products showed overall good quality, reliable values at global scale and consistent with ground references and equivalent regional (Europe) medium-resolution and high-resolution satellite sensors estimates, except for the non-expected values over few desert areas. The spatial precision over homogeneous areas covering different biomes is good, with STD and MAD

typically lower than 4% and 9% for SOSD and EOSD. Equatorial regions displayed larger spatial variability which could be either due to natural variability in the area or an impact of the noise in the input data due to the frequent cloud cover over these regions. Large spatial variability was found for productivity SPROD and TPROD layers (STD and MAD typically between 10% and 25%). The variations of SOSD and EOSD per latitude (latitudinal transect from Norway to South Africa) showed the expected trend and consistent values with estimates from VIIRS (from a previous year) except over the Sahara and scrubs areas in the south.

CGLS OLCI LSP provided good accuracy for SOSD with bias typically ranging between one week with the phenophases related to start of the season in PEP725, TEMPO and NEON, and larger negative bias with phenophases related to more developed vegetation (e.g., >50% leaf developed). For EOSD, the bias is typically negative, and more similar to colouration stages or beginning of dormancy that to the fall of leaves. The percentages of CGLS OLCI LSP cases within goal requirements are typically ranging between 10% and 20% for SOSD (25%-40% for threshold) and between 15% and 20% for EOSD (30%-45% for threshold).

GCLS OLCI LSP tends to provide shorter seasons as compared with near-ground-based Phenocam and satellite-based datasets over Europe. CGLS OLCI LSP showed typically delayed SOSD than MR-VPP (B=18 days) but better accuracy with ground observations, HR-VPP (B=14 days) and PhenoCam (B=12 days). In case of EOSD, CGLS OLCI showed similar values than MR-VPP and HR-VPP (B of -2 and 4 days) and anticipated EOSD than PhenoCam.

The comparison of productivity layers (SPROD and TPROD) showed high correlations and higher productivity (positive bias around 25%-30%) than MR-VPP and HR-VPP. Comparisons of CGLS TPROD and SPROD with other productivity GPP global products showed consistent values: $R \geq 0.76$ with CGLS OLCI GPP and $R \geq 0.67$ with MODIS GPP.

5 PRODUCT USAGE

CGLOPS LSP product offers invaluable insights across various domains. It enables continuous ecosystem monitoring, providing critical data for assessing impacts of environmental stresses like drought, identifying disturbances like deforestation and invasive species encroachment, facilitating timely decisions and interventions. Phenological shifts serve as sensitive indicators of climate change impacts, aiding in climate studies and predicting ecosystem vulnerability. In assessing drought impacts, phenology data help quantify the severity of water stress on vegetation, informing resource management and mitigation strategies. For land use planning, understanding phenology guides sustainable management practices by assessing land cover changes and their ecological consequences. In crop yield estimation, phenology data contribute to optimizing agricultural practices, predicting yield potential, and ensuring food security. In biodiversity conservation, phenology data aid in understanding ecosystem responses to stressors and climate change, informing effective conservation strategies and habitat protection efforts. In summary, the application of LSP enhances our understanding of ecosystem dynamics, supporting informed decision-making across conservation, climate adaptation, drought management, land use, and agricultural sectors.

The potential application of LSP is exemplified in the following figures. Figure 9 demonstrates the total productivity (TPROD) in 2023 over the Eastern Hemisphere (Eurasia and Africa) and the Western Hemisphere (South and North Americas). Highest total productivity is observed in three tropical rainforest regions: the Amazon, Congo Basin, and Southeast Asia. Intensively managed croplands in the eastern US and central-eastern Europe also exhibit high productivity. Boreal forests showed moderate total productivity. The peak growing season demonstrates monotony in northern and southern temperate zones, with the majority occurring in June-July and January-February, respectively. Tropical zones demonstrated more diverse peak growing seasons (Figure 10). Figure 11 shows LSP over two contrasting climate zones: Sahelian desert croplands and Amazon rainforest zones. The cropland productivity in 2023 matched well with the land green patten shown in Sentinel-2 Red-Green-Blue (RGB) color image, while the varied levels of productivities across different cropland parcels can be seen in LSP TPROD image (Figure 11b), not shown in the Sentinel-2 RGB image (Figure 11a). The Amazon rainforests host the highest global biodiversity, which is also evident in the start and end of growing season dates (SOSD in Figure 11c, and EOSD in Figure 11d). The SOSD and EOSD can occur in any month of the year, depending on the environmental drivers and limitations such as solar radiation and rainfall, as well as resource-competitions within plant communities: understorey vegetation or above-storey canopy. CGLOPS LSP will provide unprecedent evidence in understanding vegetation phenology variations and environmental controls across the entire Amazon rainforest region.

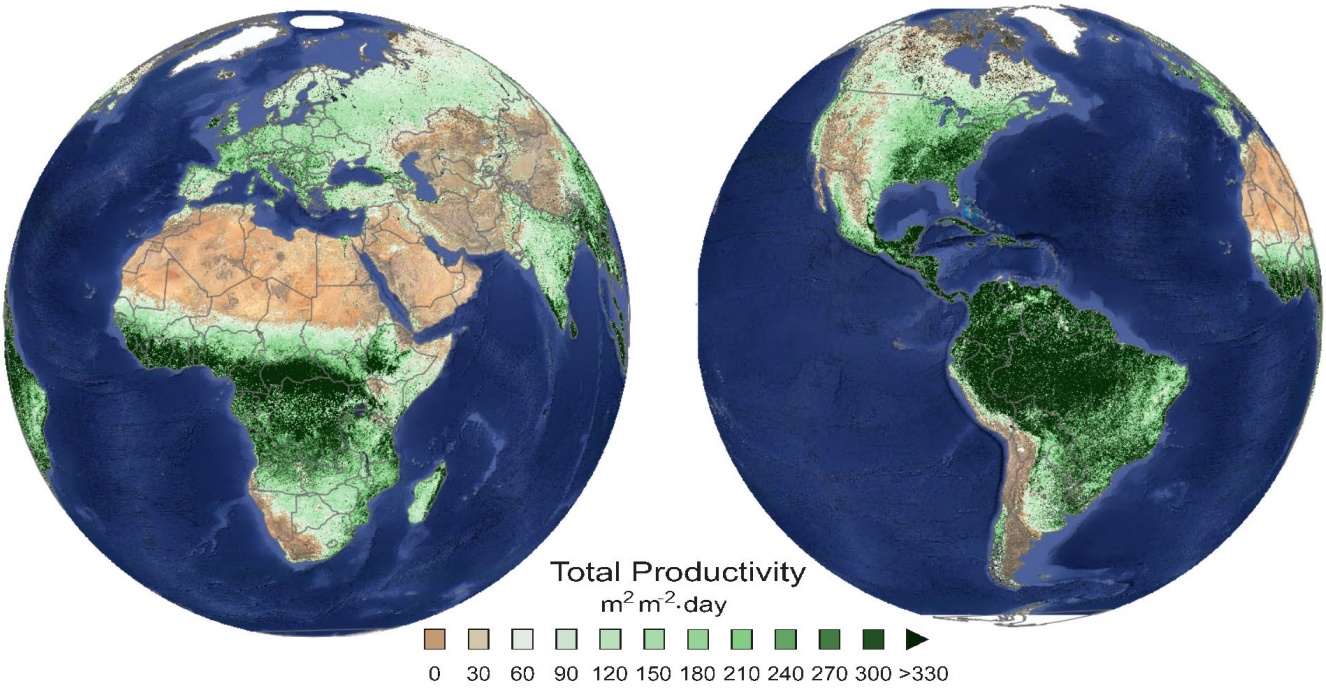


Figure 9. Global terrestrial vegetation production in 2023. Three rainforests—the Amazon rainforest, the Congo Basin rainforest, and the Southeast Asian rainforest—have the highest productivity among the world's land cover types. Agricultural lands in the eastern United States and central-eastern Europe also exhibit high productivity due to intensive farming practices.

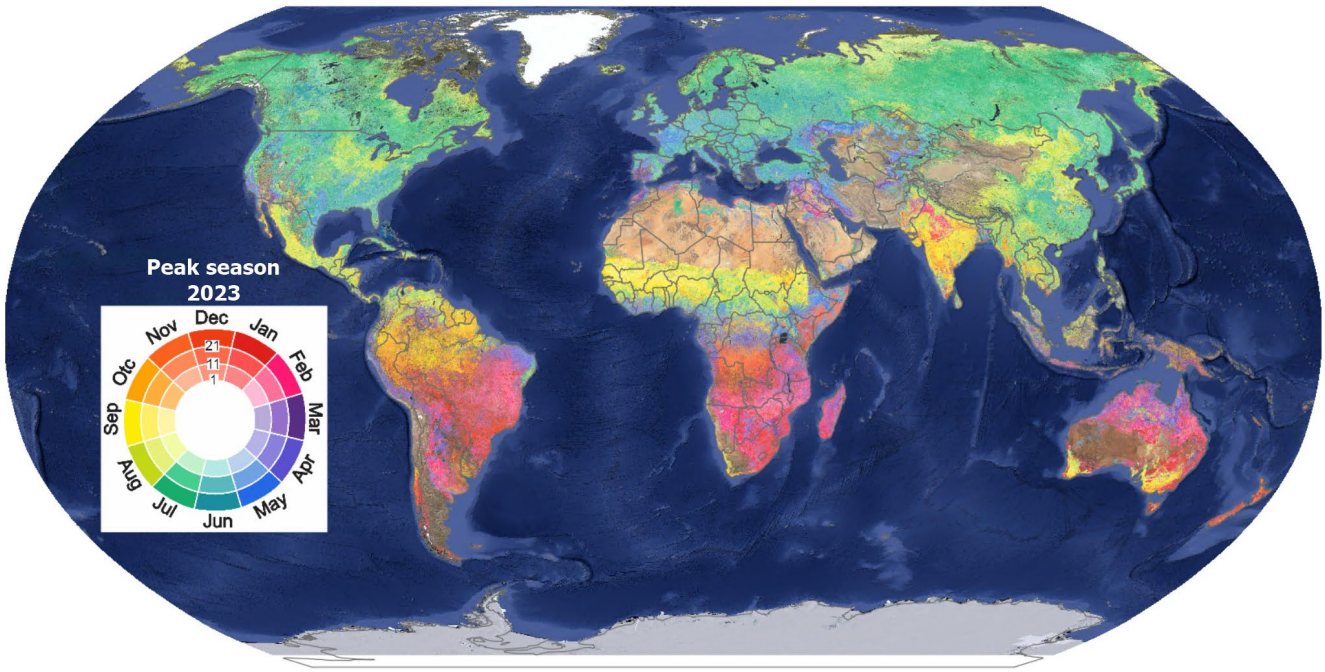


Figure 10. Map of peak growing season dates in 2023, demonstrating the gradient of phenological biodiversity at a global scale: diversity decreasing poleward from the equator.

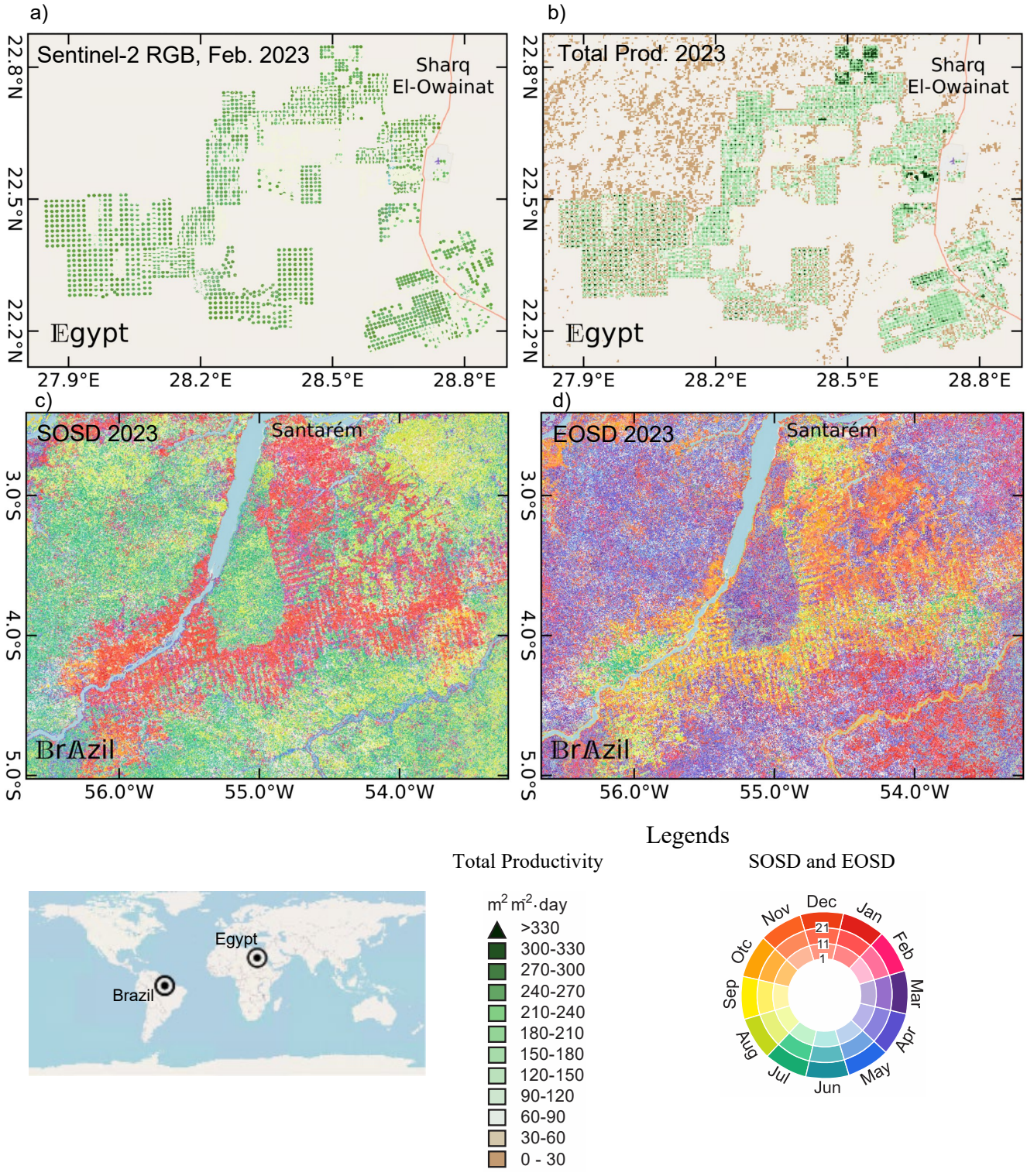


Figure 11. Local scale studies. (a, b) Agricultural land total productivity (TPROD) in African dessert. (c, d) Phenological biodiversity in Amazon rainforest regions, evident by the diverse start of the season dates (SOSD) and the end of the season dates (EOSD).

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7 ANNEX: USERS REQUIREMENTS

According to the applicable document [AD1], the user's requirements relevant for the Land Surface Phenology (LSP) are described below.

7.1 SPECIFIC TECHNICAL DETAILS AND REQUIREMENTS

PRODUCT SPECIFICATION	
Geometric Properties:	
Spatial resolution	300 m
Geolocation precision	Better than 0.5 pixels
Coordinate position	Centre of the pixel
Geodetic datum	WGS84
Geographic projection	Regular latitude/longitude grid
Geographic coverage	Global
LSP – LAND SURFACE PHENOLOGY	
LSP indicates the timing, date of occurrence, and corresponding value of the vegetation parameter, of key phases of the vegetation development cycle; Phenological metrics related to the start, end of growing season, length of the season and maximum development, seasonal total and partly integrals (as productivity parameters), as well as growth and senescence rates, intra-annual coefficient of variation, etc.	
Time span	Yearly
Timeliness	after one month of last year observation
Uncertainty (2-sigma)	Dates related metrics: Threshold: 10 days, Goal: 5 days
Stability (per decade, i.e. 10 years)	Threshold: 10% Goal: 5%

7.2 FURTHER REQUIREMENTS

7.2.1 Output product composition

Products may contain various information layers and ancillary information – the base reference for product packages are the operational products as on 01.03.2023.

7.2.2 Data Structure

Data coding[1] shall be compatible with the Global Land products as on 01.03.2023 and/or follow the INSPIRE specifications, where applicable.
Ancillary information shall be as currently used and include at least the following:

- The number of measurements per pixel used to generate any synthesis product
- The per-pixel date of the individual measurement or the start-end dates of the period covered
- Quality indicators, with explicit per-pixel identification of the cause of anomalous parameter result.

The product naming and filename conventions that are used in the Copernicus Global Land component production as on 01.03.2023 shall be followed. This may be adapted for complete product collections upon agreement with the contracting authority during the Framework.

7.2.3 Data format

To ensure interoperability with the current Global Land component (operational product data formats and archive data formats) and other Copernicus services, all datasets will be available in NETCDF. Additional format such as Cloud Optimized Geotiff (COG) or ZAR format can be proposed for production or could be requested by the contracting authority during the framework contract.

7.2.4 Uncertainties and Validation

Uncertainties indicated in the product specifications above follow the threshold and goal proposed by GCOS#245.

Uncertainties estimates should account for the error propagation uncertainty coming from input data though the retrieval algorithms as during the contract period, ESA plans to imbed uncertainties in the Sentinel-3 ground segment products and, for Sentinel-2, there is an offline tool to determine Evel 1 uncertainties; these can be used for propagation in the production chain.

Validation of the products shall conform to at least the CEOS LPV standards. Wherever appropriate the bio-geophysical variables shall be validated and compared to CEOS CAL/VAL data sets and/or Ground-Based Observations for Validation (GBOV) of Copernicus Global Land component when biophysical parameters are available.

7.2.5 Input data

Copernicus sentinel data are available from https://dataspace.copernicus.eu

Bio-geophysical variable products should be based on common base reflectance data:

[1] Data coding is the provision for the number of bits, range of values, usage of reserved values, content of status map, conversion to physical values formulas.

- Sentinel 3: Derived Top-of-Canopy reflectance may be brokered or produced as it is the case on 01.03.2023 under the CGLOPS contracts;
- Sentinel 2: the Global Land Sentinel-2 Global Mosaic (S2GM)[2] component provides temporal mosaic of surface spectral bands that can be brokered and/or directly used.

Up to and including 2019, products archives of the Global Land component have been based on SPOT VGT, Proba-V, ENVISAT, MODIS, TOPEX/Poseidon, Jason-1, Jason-2, Jason-3, datasets, which are available through the https://dataspace.copernicus.eu.

LST and SWI can be based on geostationary and other satellite data.

Ancillary satellite data that is purchased through Copernicus and put at disposal of the Services is available through the Data Warehouse and will become available on the Copernicus Dataspace Ecosystem.

Ancillary data sets, other than satellite imagery described above, that might be required shall be the responsibility of the contractor.

7.2.6 Product delivery

Products shall be delivered to the Copernicus Land Component dissemination.
The Copernicus Data Space Ecosystem infrastructure will be used to provide access to the final map products.

[2] https://land.copernicus.eu/en/products/global-image-mosaic